UM-3 model

The universe model UM-3: the author's spacetime of eight dimensions, the fields and where they live, the complete action, and every parameter

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Model of UM-3, 2026-10-24 Repository once-ere/Dirac_claude, folder universe_model_3

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1 How to read this document

1.1 What this document contains

This document defines the universe model UM-3, the revision of 2026-10-24 ordered by the author: the spacetime and its metric, the fields and where they live, the complete action from which every field equation follows, and every parameter with its value and its origin. It is self-contained: every equation is written out (the terms of the action carry the tags IN-1 to IN-16 and the metric the tag FE-27, the same tags as in every UM-3 document), and every symbol of every display equation is defined right after it.

1.2 The status labels

Every equation and every parameter carries one of the labels **AUTHOR** (from the author: his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb` and his order of 2026-10-24), **RECORD** (from the record of this repository: the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment), **STANDARD** (textbook physics or a published measurement), **ASSUMED** (a choice of UM-3 without derivation), **INVENTED** (a new interaction of UM-3: the portals of dirac16complex), **DERIVED** (computed and verified in UM-3), **ESTIMATE** (an order of magnitude) and **OPEN** (not settled).

1.3 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, x_1, x_2, x_3 ordinary space, x_4 the time, and x_5, x_6, x_7 three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order. Units are natural, $\hbar = c = 1$. A prime on a function of the time x_4 alone is its derivative with respect to x_4.

2 The spacetime

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\
&\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\
g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.

- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric, with his two functions of the time $a_4(x_4)$ and $b_4(x_4)$ (b_4 is never identically 1, by the author's order) and his constant H. The metric depends on x_0 only through $z = 6Hx_0$; one period $0 < z < \pi$ of x_0 is, in the proper coordinate σ of the hidden direction, the whole line $-\infty < \sigma < \infty$ (DERIVED). In that coordinate ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$, which has a kink at $\sigma = 0$ ($z = \pi/2$): this seven-dimensional surface is the fold. The rest of spacetime is the bulk. Status: AUTHOR (the metric), DERIVED (the chart).

3 The fields and where they live

field	symbol	what it is	lives in	couples to
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions, with the author's Lovelock weights	bulk and fold	every field
dirac16complex	Ψ	the author's 16-component complex spinor field, with left- and right-handed halves Ψ_L, Ψ_R	the bulk	gravity (canonical spin connection), itself, every Standard-Model field at the fold; only Ψ_L to the Higgs doublet
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, $\lambda > 0$, $\mu^2 < 0$, expanded about the ad hoc vacuum $(\mathcal{A}, v + h)/\sqrt{2}$	the fold	gravity, W, B, the fermions, Ψ_L
lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, three generations	the fold	gravity, W, B, Φ, Ψ_L, Ψ_R
right-handed charged leptons	$e_{\mathrm{R}i}$	electron, muon, tau	the fold	gravity, B, Φ, Ψ_L, Ψ_R
right-handed neutrinos	$\nu_{\mathrm{R}i}$	three gauge singlets: Dirac neutrino masses (accepted by the author, 2026-10-24 17:32)	the fold	gravity, Φ, Ψ_L, Ψ_R
electroweak gauge fields	W^1, W^2, W^3, B	the gauge fields of $SU(2)_\mathrm{L}$ and $U(1)_Y$	the fold	gravity, Φ, the fermions, Ψ_L, Ψ_R
quarks	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	three generations of doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_L, Ψ_R
gluons	G^A	the gauge field of $SU(3)_\mathrm{C}$	the fold	gravity, quarks, Ψ_L, Ψ_R
stabiliser	$\mathcal{T}_\mathrm{stab}$	a medium that keeps the late-time bulk static	the bulk	gravity
fold fluid	$\mathcal{T}_\mathrm{ff}$	the part of the thin source of the fold not supplied by the other fields	the fold	gravity
induced gravity	κ_4	an Einstein-Hilbert term of the fold	the fold	gravity

The Standard Model lives on the fold, because in the bulk every species would move with its own limiting speed (the warp averaged over its profile) and three neutrino flavours with three profiles would oscillate with phases that measurements exclude; on the fold every field has the one light cone of the fold (the limiting speeds are DERIVED; the place is ASSUMED on that basis). dirac16complex, the author's field of eight dimensions, lives in the bulk and couples to the Standard Model at the fold. Status: AUTHOR (the list of fields), ASSUMED (the additions and the places).

4 The action

All field equations of UM-3 follow from one action by varying it with respect to every field.

4.1 The sum of all terms

$$
\begin{aligned}
S_{\mathrm{tot}} &= S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_{\Psi} + S_{\mathrm{fold}},\\
S_{\mathrm{fold}} &= \int \mathrm{d}^8X\,\sqrt{|g|}\,\delta_{\mathrm{D}}(\sigma)\,\frac{1}{V_*}\left(\mathcal{L}_{\mathrm{IG}} + \mathcal{L}_{\mathrm{ff}} + \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} + \mathcal{L}_{\mathrm{P}}\right)
\end{aligned}
\tag{IN-1}
$$

Symbols in (IN-1).

- S_{tot}: The action of UM-3 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- S_{stab}: The action of the stabiliser of the bulk (a medium specified by its energy-momentum tensor $\mathcal{T}_{\mathrm{stab}}$). Unit: dimensionless. Status: ASSUMED.
- S_{Ψ}: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{fold}: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the induced gravity of the fold, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_{\mathrm{D}}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_{\mathrm{IG}}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\mathrm{ff}}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_{\mathrm{ff}}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has four parts: Einstein-Lovelock gravity in eight dimensions, the stabiliser of the bulk, dirac16complex in the bulk, and the fold, whose terms are confined to $\sigma = 0$ by the Dirac delta of the proper hidden coordinate. Status: ASSUMED.

4.2 Gravity

$$
\begin{aligned}
S_{\mathrm{EL}} &= \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\left(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\right), \qquad \kappa = 8\pi G_8,\\
\alpha_{(p)} &= w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}, \qquad \Lambda = -10H^2
\end{aligned}
\tag{IN-2}
$$

Symbols in (IN-2).

- $S_{\rm EL}$: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\rm N} = 10$ in $0 = -\Lambda_{\rm N} + \sum_p H^{-2p} w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]). Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \end{aligned} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\rm L} = 1$ for $X = \rm L$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: 1/length^2. Status: STANDARD.

Einstein-Lovelock gravity with the author's weights and cosmological constant (his notebook, In[134]): the Einstein-

Hilbert, the Gauss-Bonnet and the cubic Lovelock densities, built from the Riemann tensor with the generalised Kronecker delta. Their variation contains no derivative of the metric higher than the second, so the field equations are of second order and a thin source at the fold produces a finite jump of first derivatives. Status: AUTHOR, STANDARD.

4.3 dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1,\ldots,x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2,\ldots$): it runs over $x_0, x_1,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

$$\begin{aligned}
&\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\,T^a,\\
&\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7,\\
&P_\mathrm{L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_\mathrm{R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_\mathrm{L} = P_\mathrm{L}\Psi, \qquad \Psi_\mathrm{R} = P_\mathrm{R}\Psi
\end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- T_a, T^a: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0,\ldots,7$; $T_0,\ldots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\ldots,7$ (frame direction (0) along σ, (1),(2),(3) along ordinary space, (4) along the time, (5),(6),(7) along the extra times); raised and lowered with η_{ab}; on the fold ($a,b\in\{1,\ldots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\ldots,x_7$ for a tensor of eight dimensions, $x_1,\ldots,x_7$ for a tensor of the fold, $x_1,\ldots,x_4$ for a four-dimensional tensor). Status: STANDARD.

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16ˆA[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Ψ_L: The left-handed half of dirac16complex, $\Psi_\mathrm{L} = P_\mathrm{L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ψ_R: The right-handed half of dirac16complex, $\Psi_\mathrm{R} = P_\mathrm{R}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\,\Omega_M, \qquad \Omega_M = \frac{1}{2}\,\varpi_{Mab}\,\Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big],$$
$$\varpi_{Mab} = \mathbf{e}_a{}^N\big(\partial_M \mathbf{e}_{bN} - \Gamma^P{}_{MN}\,\mathbf{e}_{bP}\big) \tag{IN-6}$$

Symbols in (IN-6).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ϖ_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

dirac16complex with the kinetic term in the symmetric form (real for the author's real matrices), the mass m_Ψ and the contact self-interaction λ_Ψ (RECORD); the author's eight real 16×16 Dirac matrices, the Dirac adjoint and the chiral projectors (AUTHOR); the canonical spin connection, which couples dirac16complex to gravity and is never identically zero for the author's metric ($\gamma^M\Omega_M=-3H\,\mathrm{sgn}(\sigma)\gamma^{(0)}+\frac{3}{2}(b_4'/b_4)\gamma^{(4)}$, DERIVED; here $\gamma^{(0)}=T^0$ and $\gamma^{(4)}=T^4$ are the author's matrices along the hidden direction and along the time, σ the proper coordinate of the hidden direction with the fold at $\sigma=0$, and sgn the sign function).

4.4 The stabiliser of the bulk

$$\mathcal{T}_{\mathrm{stab}}{}^{a}{}_{b}=\mathrm{diag}\big(p_\mathrm{s},p_\mathrm{s},p_\mathrm{s},p_\mathrm{s},-\rho_\mathrm{s},p_\mathrm{s},p_\mathrm{s},p_\mathrm{s}\big),\qquad \kappa\rho_\mathrm{s}=-\frac{H^2}{16},\qquad \kappa p_\mathrm{s}=-\frac{5H^2}{16} \tag{IN-7}$$

Symbols in (IN-7).

- $\mathcal{T}_{\mathrm{stab}}{}^{a}{}_{b}$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0,1,\ldots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a,b\in\{1,\ldots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\ldots,x_7$ for a tensor of eight dimensions, $x_1,\ldots,x_7$ for a tensor of the fold, $x_1,\ldots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_s: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_\mathrm{s}=-5H^2/16$. Status: ASSUMED.
- ρ_s: The energy density of the stabiliser. Unit: GeV^8. Value: $\kappa\rho_\mathrm{s}=-H^2/16$. Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda=-10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The medium of the bulk that the late universe needs: with the author's weights the bulk of a slowly expanding universe is not a vacuum solution, and its static state requires this energy-momentum tensor, with a negative energy density $\rho_\mathrm{s}=-H^2/(16\kappa)$ and the pressure $p_\mathrm{s}=5\rho_\mathrm{s}$; it violates the null energy condition. Status: ASSUMED (its existence), DERIVED (its value).

4.5 The fold

$$\int \mathrm{d}^8X\,\sqrt{|g|}\,\delta_\mathrm{D}(\sigma)\,\frac{1}{V_*}\,\mathbb{F}=\int \mathrm{d}^4x\,\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathbb{F}\big|_{\sigma=0},\qquad \mathrm{d}^4x=\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4,$$
$$\hat{h}_{\mu\nu}=\mathrm{diag}\big(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1\big),\qquad \sqrt{-\hat{h}}=\mathrm{e}^{3a_4},\qquad \mathcal{Z}=\frac{V_\mathrm{t}\big(\mathrm{e}^{-a_4}b_4\big)^3}{V_*},\qquad V_\mathrm{t}=\int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7 \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_D: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_\mathrm{D}(\sigma)F(\sigma)=F(0)$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5,x_6,x_7 on the fold in the late universe, $V_*=V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.

- $\mathbb{F}$: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

$$\mathcal{L}_\mathrm{IG} = \frac{\hat{R}}{2\kappa_4} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_\mathrm{IG}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_\mathrm{ind}$; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_4/(1 + 5\epsilon_\mathrm{r}/6)$ (the Friedmann equation of the fold to first order in ϵ_r). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_\mathrm{N} = 8\pi/M_\mathrm{Pl}^2$. Status: ASSUMED.

$$\mathcal{T}_\mathrm{ff}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{ff}, p_\mathrm{ff}, p_\mathrm{ff}, -\rho_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_\mathrm{ff}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.

- p_{ff}: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- q_{ff}: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.

The fold: the reduction of the terms of the fold to four-dimensional fields with the volume V_* of the extra times and the factor $\mathcal{Z}$ (IN-8); the induced gravity of the fold, which makes four-dimensional gravity general relativity (IN-9), because the gravitons of the bulk move 2.74 times faster than light at the static block (the late-time state of the bulk, in which a_4 and b_4 are constant) and have no time-kinetic term at the author's vacuum (the history $a_4 = Hx_4$ with constant b_4, which needs no sources in the bulk); the fold fluid, the thin source that the kink of the warp requires (IN-10). Status: ASSUMED, DERIVED (the reduction).

4.6 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_{\mathrm{H}} = -\hat{h}^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi - V(\Phi), \qquad V(\Phi) = \mu^2\,\Phi^\dagger\Phi + \lambda\big(\Phi^\dagger\Phi\big)^2, \qquad \lambda > 0, \quad \mu^2 < 0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\mathcal{A}(x_4)\\ v + h\end{pmatrix}, \qquad \mathcal{A}(x_4) = v_{\mathcal{A}}\,\mathrm{e}^{-2a_{40}x_4}, \qquad a_{40} > 0, \quad x_4 \geq 0, \qquad v^2 = -\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$ of the muon lifetime (Particle Data Group). Status: STANDARD.
- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.

- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc = Exp$[-2\,a_{40}\,x_4]$ has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-3 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The Higgs doublet with the author's potential, expanded about the author's ad hoc vacuum, whose upper component almightyAdHoc decays as $\mathrm{e}^{-2a_{40}x_4}$ (the author writes Exp$[-2a_{40}x_4]$; UM-3 gives it the amplitude $v_{\mathcal{A}}$ so that it has the dimension of the Higgs field). Status: AUTHOR.

4.7 The gauge fields, the quarks and the leptons

$$
\begin{aligned}
D_\mu &= \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G^A_\mu\, t^A - \mathrm{i}g_2\, W^I_\mu\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu, \\
\mathcal{L}_{\text{gauge}} &= -\frac{1}{4}\, G^A_{\mu\nu} G^{A\mu\nu} - \frac{1}{4}\, W^I_{\mu\nu} W^{I\mu\nu} - \frac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \\
G^A_{\mu\nu} &= \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_3\, f^{ABC} G^B_\mu G^C_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_2\, \epsilon^{IJK} W^J_\mu W^K_\nu, \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu
\end{aligned}
\tag{IN-13}
$$

Symbols in (IN-13).

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- $G^A_\mu,\ G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A = \lambda^A_{\mathrm{GM}}/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- $W^I_\mu,\ W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value:

$g_1(m_Z) = 0.3574$. Status: STANDARD.

- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, −1/3, −1/2, −1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.

$$\mathcal{L}_\mathrm{f} = \sum_\psi \mathrm{i}\,\bar\psi\,\hat\gamma^\mu D_\mu \psi, \qquad \psi \in \{Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}\},$$
$$\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}, \qquad \bar\psi = \psi^\dagger\beta, \qquad \beta = \hat\gamma^{(4)}, \qquad \hat\gamma_5 = \mathrm{i}\,\hat\gamma^{(4)}\hat\gamma^{(1)}\hat\gamma^{(2)}\hat\gamma^{(3)},$$
$$\hat P_\mathrm{L} = \frac{\mathbf{1} - \hat\gamma_5}{2}, \quad \hat P_\mathrm{R} = \frac{\mathbf{1} + \hat\gamma_5}{2}, \qquad \hat P_\mathrm{L} Q = Q, \quad \hat P_\mathrm{L}\ell = \ell, \quad \hat P_\mathrm{R} u_\mathrm{R} = u_\mathrm{R}, \quad \hat P_\mathrm{R} d_\mathrm{R} = d_\mathrm{R}, \quad \hat P_\mathrm{R} e_\mathrm{R} = e_\mathrm{R}, \quad \hat P_\mathrm{R}\nu_\mathrm{R} = \nu_\mathrm{R} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu \hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge −1/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge −1/2). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge −1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta = \hat\gamma^{(4)}$ (Hermitian, $\beta^2 = \mathbf{1}$). Unit: dimensionless.

Status: STANDARD.

- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5 = \mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2 = \mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{L}$: The four-dimensional left-handed projector, $\hat{P}_\mathrm{L} = (\mathbf{1} - \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{R}$: The four-dimensional right-handed projector, $\hat{P}_\mathrm{R} = (\mathbf{1} + \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

$$\mathcal{L}_\mathrm{Y} = -\big(\bar{Q}\,\mathcal{Y}^u\,\tilde{\Phi}\,u_\mathrm{R} + \bar{Q}\,\mathcal{Y}^d\,\Phi\,d_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^e\,\Phi\,e_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^\nu\,\tilde{\Phi}\,\nu_\mathrm{R}\big) + \mathrm{h.c.}, \qquad \tilde{\Phi} = \mathrm{i}\tau^2\Phi^* \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The gauge fields of $SU(3)_\mathrm{C} \times SU(2)_\mathrm{L} \times U(1)_Y$ on the fold (IN-13), the three generations of quarks and leptons with three right-handed neutrinos (IN-14), and their Yukawa couplings to the Higgs doublet (IN-15). With the right-handed neutrinos the gauge anomalies still cancel (DERIVED, `um3_fields.py`), and the neutrino Yukawa couplings give the neutrinos Dirac masses, whose values are chosen to reproduce the measured mass differences with the lightest mass zero (ASSUMED). Status: STANDARD (UM-3 adds the right-handed neutrinos).

4.8 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_\mathrm{P} = & -c_\Phi\,\big(\Phi^\dagger\Phi\big)\,\bar{\Psi}P_\mathrm{L}\Psi - \sum_X \Big[\frac{1}{4}\big(c_X^G\,G^A_{\mu\nu}G^{A\mu\nu} + c_X^W\,W^I_{\mu\nu}W^{I\mu\nu} + c_X^B\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar{\Psi}P_X\Psi \\ & + \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_f c_X^f\,J_f^\mu\,j_{X\mu}\Big], \\ \mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar{\Psi}\,\Sigma^{\mu\nu}P_X\Psi, & \qquad j_X^\mu = \mathrm{i}\big(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi\big), \qquad J_f^\mu = \bar{\psi}_f\,\hat{\gamma}^\mu\psi_f\end{aligned} \tag{IN-16}$$

Symbols in (IN-16).

- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f\hat{\gamma}^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-3: every Standard-Model field couples at the fold to the left-handed and to the right-handed half of dirac16complex, each with its own coupling, through gauge-invariant operators: the Higgs portal (left-handed half only), the gauge-kinetic portals of the gluons, W and B, a dipole portal to B, and current portals between the fermion-number current of every multiplet of the Standard Model and the convective currents of the two halves. Each is invariant under the gauge group of the Standard Model and contains dirac16complex only in bilinears, so the charge of dirac16complex is conserved and its lightest states cannot decay into the Standard Model alone; whether they are stable against the negative-energy states of its indefinite charge is open. Status: INVENTED (the portals), OPEN (the stability).

5 The parameters

parameter	meaning	value	status
H	the author's constant: the period $\pi/(6H)$ of x_0, the warp, the Lovelock weights	free; the bound states of dirac16complex need $H \ll q_\mathrm{b}$, where q_b is the inverse width of their profile $\mathrm{e}^{-q_\mathrm{b}\lvert\sigma\rvert}$ along the hidden direction	AUTHOR
$a_4(x_4)$, $b_4(x_4)$	the author's functions of the time	the author's vacuum $a_4 = Hx_4$ with constant b_4 (primordial); frozen extra times $\mathrm{e}^{-a_4}b_4 =$ constant (late)	AUTHOR, DERIVED
$\alpha_{(p)}$, Λ	Lovelock weights and cosmological constant of the bulk	$\alpha_{(p)} = H^{2-2p}w_{(p)}$ with $w = (1, 1/32, 1/1152)$, $\Lambda = -10H^2$	AUTHOR
$\kappa = 8\pi G_8$	the gravitational coupling of the bulk	free	ASSUMED
κ_4	the coupling of the induced gravity of the fold	$8\pi G_\mathrm{N}(1 + 5\epsilon_\mathrm{r}/6)$, with Newton's constant G_N	ASSUMED, DERIVED
V_*	the physical volume of the extra times at the fold	free	ASSUMED
$\epsilon_\mathrm{r} = \kappa_4 V_*/(\kappa H)$	the ratio of bulk to fold gravity	$\lesssim 10^{-15}$: if a fraction ϵ_r of the gravitational waves of the fold travelled with the bulk gravitons (speed squared 15/2 times that of light), their speed would differ from that of light by about $3\epsilon_\mathrm{r}$; the waves of the neutron-star merger GW170817 arrived with its light to about 10^{-15}	DERIVED (the definition), ESTIMATE (the bound)
ρ_s, p_s	the stabiliser	$-H^2/(16\kappa)$, $5\rho_\mathrm{s}$	DERIVED
ρ_ff, p_ff, q_ff	the fold fluid (energy density, pressure along ordinary space, pressure along the extra times)	$\rho_\mathrm{ff} - 31HV_*/(4\kappa) = \rho_\Lambda =$ $(2.2\ \mathrm{meV})^4$, the vacuum energy of the fold (the dark energy)	ASSUMED
m_Ψ	the bulk mass of dirac16complex	free	RECORD
λ_Ψ	the self-coupling of dirac16complex	free	RECORD
c_Φ, c_X^G, c_X^W, c_X^B, d_X, c_X^f	the portal couplings ($X =$ L, R)	free; direct detection and the effective number of neutrino species N_eff (the energy density of the relativistic species besides the photons, in units of one neutrino species) constrain them	INVENTED
v, λ, μ^2	the Higgs vacuum and potential	$v = 246.22$ GeV, $\lambda = 0.129$, $\mu^2 = -\lambda v^2$	AUTHOR, STANDARD

parameter	meaning	value	status
$v_{\mathcal{A}}$, a_{40}	amplitude and rate of the ad hoc vacuum $\mathcal{A} = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$	the author's amplitude is 1 in the units of his notebook; UM-3 keeps $v_{\mathcal{A}}$ free (the examples take v and $10v$); $a_{40} > 0$; nucleosynthesis needs $a_{40} > \ln(v_{\mathcal{A}}/(0.1v))/(2\ \mathrm{s})$	AUTHOR, ASSUMED (a free amplitude)
g_1, g_2, g_3	the gauge couplings	the measured values	STANDARD
$\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$	the Yukawa matrices	the measured masses and mixings (Particle Data Group); neutrinos (the global fit NuFIT 5.2 of the oscillation data, November 2022, normal ordering, with the Super-Kamiokande atmospheric data): the differences of the squared masses of the three mass states $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2 and $\Delta m^2_{31} = 2.511 \times 10^{-3}$ eV^2, the mixing angles $\theta_{12} = 33.40^\circ$, $\theta_{13} = 8.54^\circ$, $\theta_{23} = 49.14^\circ$; the lightest mass 0 and the Dirac phase 0	STANDARD, ASSUMED (the lightest mass and the phase)
A_s, n_S, τ_reion	the primordial spectrum and the optical depth	Planck 2018: $\ln(10^{10} A_\mathrm{s}) = 3.044$, $n_\mathrm{S} = 0.9649$, $\tau_\mathrm{reion} = 0.0544$	ASSUMED